

Sampling & Good Measurement (Reliability & Validity)

Who do we study—and how do we know our measures are any good?

WEEK 5 • DAY 2

Two questions for today

WHO do we measure?

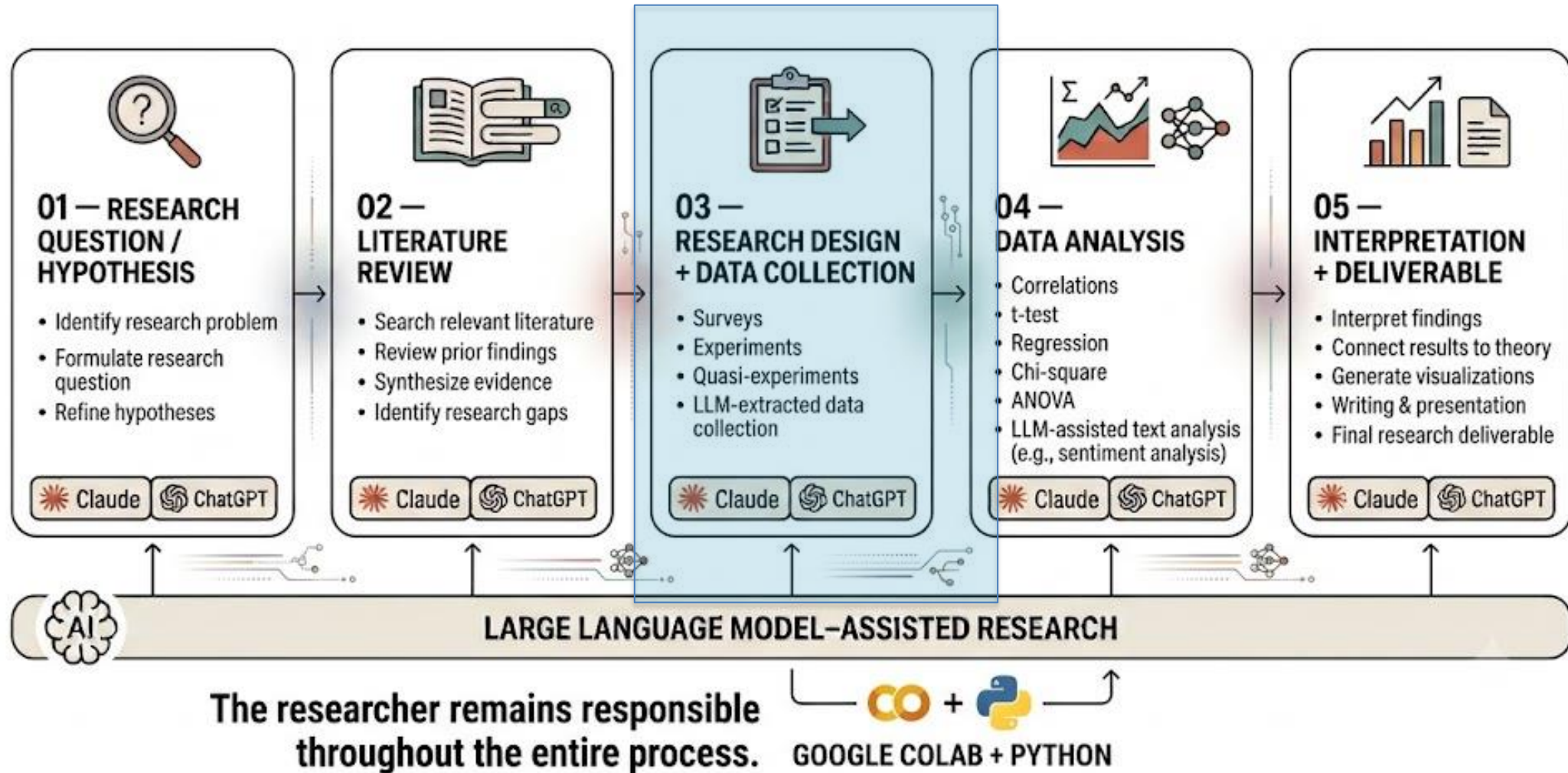
Sampling: choosing the people, posts, organizations, or other cases that will provide our data.

HOW WELL do we measure?

Measurement quality: deciding whether our questions or scales consistently and appropriately represent the concept.

Connect to Day 1

Research Methods in the Age of AI



What is sampling?

"RQ: How much do college students trust news they encounter on TikTok?"

- We want to know something about a large group: college students.
- But we cannot realistically ask every college student.
- So we study a smaller group and use that evidence to learn about the larger group.

This is the basic problem that sampling solves.

Population vs. Sample

POPULATION

The full group you want your study to say something about.

Example: all undergraduate students at a university.

SAMPLE

The smaller group you actually collect data from.

Example: 300 undergraduate students who complete your survey.

The population must match your research question

RQ: How do USC undergraduates use TikTok for news?

Population: USC undergraduate students.

RQ: How do U.S. sports journalists use AI tools?

Population: sports journalists working in the United States.

RQ: How do Gen Z consumers respond to influencer apologies?

Population: the Gen Z consumers your study intends to understand.

Ask: WHO do I ultimately want to understand?

What is a sampling frame?

- A sampling frame is the list or source you can actually use to identify possible participants.
- Population: all students enrolled at a university.
- **Possible sampling frame:** the university student directory or registrar list.
- Population: U.S. sports journalists.
- **Possible sampling frame:** a professional association membership list.

The sampling frame may not perfectly cover the population.

Population = who you care about. Sampling frame = who you can actually reach/select from.

Two broad families of sampling

PROBABILITY SAMPLING

Selection involves a known random process. Members of the population have a known chance of being selected.

NONPROBABILITY SAMPLING

Selection is not based on a known random process. Participants may be recruited because they are accessible or meet specific criteria.

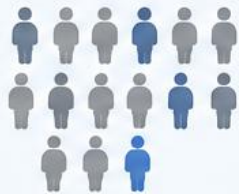
Probability Sampling

Everyone in the population has a known, non-zero chance of being selected.

- Based on random selection
- Allows statistical generalization to the population (if done well)
- Usually requires a sampling frame (e.g., a complete list)
- More time, cost, and planning

Common types

Simple Random Sampling



Every person has an equal chance of being selected.

Stratified Sampling



Divide the population into subgroups (e.g., year in school), then randomly sample from each group.

Cluster Sampling



Randomly select groups (e.g., classes or schools), then survey everyone in those groups or a random sample within them.

Non-Probability Sampling

Not everyone in the population has a known chance of being selected.

- Based on convenience or researcher judgment
- Does not allow statistical generalization to the population
- Often used when a sampling frame is not available or the population is hard to reach
- Faster, cheaper, and more flexible, but higher risk of bias

Common types

Convenience Sampling



Survey people who are easiest to reach (e.g., students in your class).

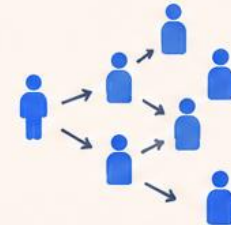
Purposive Sampling

Only people who meet specific criteria



Intentionally select people with particular characteristics (e.g., social media influencers).

Snowball Sampling



Initial participants recruit others (useful for hard-to-reach populations, e.g., LGBTQ+ communities).



Key takeaway: Probability sampling supports generalizable results, while non-probability sampling is useful when a random sample is not possible, but it comes with higher risk of bias.

Probability Sampling:

1. Simple random sampling

Imagine we have a complete list of 10,000 students.

We use a random process to select 500 names.

Each student has a known chance of selection.

This helps reduce systematic selection by the researcher.

Simple random sampling = randomly select people from a sampling frame.

2.Stratified random sampling

Sometimes we want important groups represented in the sample.

Example: divide students into first-year, sophomore, junior, and senior groups.

Then randomly sample within each group.

This can help ensure that smaller but important groups are represented.

Stratify = divide into meaningful groups first → randomly sample within each group.

Nonprobability Sampling:

1. Convenience sampling

Recruit people who are easy to access.

Example: send a survey to students in your own classes.

Example: post a survey link to your Instagram followers.

Convenience samples are fast and inexpensive.

But the people easiest to reach may differ from the population you want to understand.

Easy to recruit $\neq$ representative of everyone.

2. Purposive sampling

Intentionally recruit people because they have characteristics or experiences relevant to the research question.

Example: recruit sports journalists who have personally used generative AI in their reporting.

Example: recruit PR professionals who have managed an organizational crisis.

Useful when your question requires a specific type of participant.

Purposive = I need participants who meet a particular criterion.

3. Snowball sampling

Start with a small number of eligible participants.

Ask them to help connect you with other eligible participants.

Example: interview independent sports journalists, who then refer other independent journalists.

Useful when the population is difficult to identify or reach.

But people often refer others who are similar to themselves.

Snowball = participants help you reach additional participants.

Sampling methods at a glance

Method	How participants are selected	Communication example	Main concern
Simple random	Randomly from a list	Random students from registrar list	Need a good sampling frame
Stratified random	Randomly within groups	Random students within each class year	Need information about strata
Convenience	Who is easy to reach	Students in your own classes	Selection bias
Purposive	People meeting criteria	PR professionals who handled a crisis	Limited generalizability
Snowball	Participants recruit others	Independent journalists refer peers	Network similarity

Which sampling method?

Choose the sampling method that best describes each recruitment plan.

1. A professor emails the survey to students in her own course.
2. A researcher randomly selects 300 names from the university student directory.
3. A researcher recruits only sports reporters who have covered athlete mental-health stories.
4. Five freelance journalists participate and each refers another freelancer.
5. Students are divided by year in school and randomly selected within each year.

Sampling bias: who might be missing?

- A sample can systematically overrepresent some people and underrepresent others.
- Instagram recruitment may miss people who rarely use Instagram.
- An online survey about news may attract people who are unusually interested in news.
- A voluntary survey about a controversial brand may attract people with especially strong opinions.

The question is not only “How many people did I recruit?” but also “Who ended up in my sample?”

A large biased sample can still be biased.

Sample size is not the same as representativeness

- A sample of 5,000 people is not automatically better than a sample of 500.
- **Sample size** affects statistical precision and what analyses are possible.
- **Sampling method** affects who the sample represents.
- Both matter—but they solve different problems.

Bigger does not automatically mean more representative.

Random sampling ≠ random assignment

RANDOM SAMPLING

WHO enters the study?

Randomly select participants from a population.

Main goal: help support generalization to the population.

RANDOM ASSIGNMENT

WHAT condition do participants enter?

Randomly place participants into experimental conditions.

Main goal: help make groups comparable for causal inference.

Sampling Activity

Design a study: A PR apology campaign



Goal: Apply what you learned about sampling to a real-world communication and PR example.

Scenario: A major coffee chain, BrewBuddies, recently faced backlash after a viral TikTok showed a barista making an insensitive remark to a customer. The company released a public apology campaign, including a CEO video, new training policies, and a social media campaign with the hashtag #BetterBrewTogether.

You want to study: How does the apology campaign affect public trust in the brand?

BREWBUDDIES

[About](#)[Our Coffee](#)[Sustainability](#)[News](#)

PRESS RELEASE

BrewBuddies Announces Next Steps After Viral Incident

March 12, 2026

We are deeply sorry for the recent incident captured in a viral video. This does not reflect our values. We are taking action, including:

- A company-wide training program on inclusivity
- A new customer feedback system
- Ongoing updates on our progress

We remain committed to creating a welcoming space for everyone. #BetterBrewTogether



1

What is the population?

Define the group you want to generalize to.

2

Give one possible sampling frame.

Where would you get a list or source of people to sample from?

3

What would convenience sampling look like?

How might you recruit participants quickly and easily?

4

What would a probability sample look like?

Which method (e.g., simple random, stratified) might be appropriate, and how would it work?

5

What should we be cautious about?

If we only recruit people who comment on BrewBuddies' Instagram post about the apology, what are the potential biases?



Think → Write → Discuss

You will have a few minutes to work individually, then discuss with a partner or small group.

Now we have participants. But what if our measurement is bad?

Suppose our variable is TRUST IN BREWBUDDIES AFTER THE APOLOGY CAMPAIGN.

Sampling asks WHO. Measurement quality asks HOW WELL.

First rule: look at how previous researchers measured it

Do not automatically invent a new “brand trust” measure for BrewBuddies.

Literature review helps us find established ways to measure our variables.

What should You look for in a paper?

WHAT did they call it?

Example: source credibility, brand trust, perceived authenticity.

HOW many items?

One question? Three items? Ten items?

WHAT were the items?

Read the actual statements participants answered.

WHAT response scale?

1–5? 1–7? Strongly disagree → strongly agree?

WHERE did it come from?

Did the authors create it or adapt an established scale?

HOW did it perform?

Did they report reliability or validity evidence?

Reliability: does the measure behave consistently?

- Reliability is about consistency of measurement.
- If several items are intended to measure the same construct, their responses should generally fit together.
- Example: someone who strongly agrees that a source is trustworthy would usually also rate that source as dependable and believable.
- One common form is internal consistency reliability.

Cronbach's alpha (α): one common reliability statistic

- Cronbach's alpha summarizes how consistently a set of items tends to work together.
- It ranges roughly from 0 to 1, with higher values generally indicating greater internal consistency.
- A value around **.70 or higher** is often used as a rough introductory guideline for acceptable internal consistency.
- But .70 is NOT a universal pass/fail rule; interpretation depends on the construct, items, sample, and purpose.
- Alpha does not tell us whether the scale measures the correct concept.

High α $\neq$ automatically a good or valid measure.

Reading reliability in a research paper

Example A

“The four-item source credibility scale demonstrated acceptable internal consistency ($\alpha = .82$).”

Interpretation: the items were fairly consistent in this sample.

Example B

“The three-item trust measure had $\alpha = .54$.”

Interpretation: the items may not be working together very consistently; investigate before using the score.

How do I find Cronbach's α in a real article?

1. Open the article

Start with a peer-reviewed article that measures a construct with multiple survey items.

2. Search within the page

Press Ctrl+F (Windows) or ⌘+F (Mac).

3. Try these words

alpha • Cronbach • reliability • internal consistency • measures

4. Check Methods / Measures

Find the construct name, number of items, response scale, and original source.

5. Check Results / tables

Record the α reported for the scale in THIS sample.

Live article hunt: find the measure and Cronbach's α

PAIR ACTIVITY • 8 min

Article: “The next hype in social media advertising: Examining virtual influencers’ brand endorsement effectiveness”

Open-access peer-reviewed article • Search within the article for “alpha” or “message credibility.”

[Open article: https://pmc.ncbi.nlm.nih.gov/articles/PMC10026852/](https://pmc.ncbi.nlm.nih.gov/articles/PMC10026852/)

1. Find the section where message credibility is measured.
2. How many items are used to measure message credibility?
3. What response scale is used?
4. What Cronbach's α is reported for message credibility?
5. Where did the measure come from? Identify the cited source(s).
6. In one sentence: what does that α tell you—and what does it NOT tell you?

Debrief: what should you have found?

MEASURE

Message credibility

ITEMS

4 items: truthfulness, believability, deception (reverse-coded), and feeling accurately informed

RESPONSE SCALE

7-point disagree → agree

RELIABILITY

Cronbach's $\alpha = .93$

SOURCE

Items were taken from prior published measures (Kim et al.; Ohanian).

INTERPRETATION

.93 indicates high internal consistency in this sample. It does NOT by itself prove validity.

Do not stop at the α : trace the measure backward

This is how you find a defensible measurement

1. Find the sentence that says the measure was “adapted from” or “taken from” another study.
2. Click that citation in the reference list.
3. Locate the original or earlier scale paper when possible.
4. Compare the original items with the adapted items.
5. Ask: Did the authors change wording, remove items, or change the response scale?
6. Record BOTH: evidence from the current study and the original source of the measure.

A scale is not “good” forever. Reliability and validity evidence can change across samples, populations, languages, and contexts.

Validity: Is our study measuring and supporting what we think it is?

Validity is not just one thing. Different validity questions examine different parts of a study.

Face validity

At first glance, do the items appear to measure the intended concept?

Example: Do “trustworthy,” “honest,” and “dependable” look like reasonable indicators of brand trust?



Internal validity

Can we reasonably attribute an observed effect to the factor we think caused it, rather than another explanation?

Especially important for experiments and quasi-experiments.



External validity

How well might the findings generalize beyond the people, setting, or conditions actually studied?

Example: Do results from USC students apply to U.S. consumers more broadly?



Ecological validity

How well does the study resemble the real-world context or behavior we care about?

Example: Is viewing a realistic CEO apology post similar to how consumers encounter PR messages in real life?

Four validity questions using the BrewBuddies study

RQ: How does BrewBuddies' apology campaign affect public trust in the brand?



1. Face validity: does the measure make sense on its face?

This is the most intuitive validity check—but it is not enough by itself.

✓ Looks relevant

Brand trust items:

- “BrewBuddies is trustworthy.”
- “BrewBuddies is dependable.”
- “BrewBuddies is honest with customers.”

? Looks questionable

Supposed brand trust items:

- “BrewBuddies has attractive stores.”
- “Its drinks look trendy.”
- “Its TikToks are entertaining.”

Important:

A measure can have good face validity and still perform poorly. Researchers need stronger evidence than “these questions look right.”

2. Internal validity: is there another explanation?

This becomes especially important when we start experiments next week.

**Apology
campaign**



**Brand
trust**

Possible alternative explanations

Maybe people already liked BrewBuddies.
Maybe they also saw positive news coverage.
Maybe different groups saw different messages.

Internal validity asks whether the study design rules out plausible competing explanations well enough to support the causal conclusion.

3. External validity: who and where can we generalize to?

Connect this directly back to sampling.

1 STUDY SAMPLE

120 USC students complete the BrewBuddies study.

2 TARGET POPULATION

All U.S. consumers who encounter the apology campaign?

3 ASK

How similar is our sample to the population we want to understand?

Sampling matters here.

A convenience sample can still produce useful evidence—but we should be careful about claiming the result represents everyone.

4. Ecological validity: does the study feel like the real world?

Think about the context in which people normally encounter communication.



LOWER ecological realism

Participants read plain apology text pasted into a survey.

No brand visuals, no social feed, no comments, no surrounding news coverage.



HIGHER ecological realism

Participants view a realistic BrewBuddies Instagram/TikTok apology post with CEO video, caption, branding, and a familiar feed layout.

But realism is not automatically “better.”

More realistic settings can introduce more uncontrolled influences. Research design often involves trade-offs.

A practical workflow for choosing a measure

1. Clearly define the variable you want to measure.
2. Search Google Scholar and relevant literature for established measures.
3. Read the Methods / Measures sections of relevant papers.
4. Trace the citation to the original measure when possible.
5. Examine the actual items and response options.
6. Look for reliability and validity evidence.
7. Ask whether the measure makes sense for YOUR population and context.

Do not choose a measure only because another paper used it.

What if I adapt an existing measure?

Researchers sometimes change wording so a measure fits a new context.

Example: change “this company” to “this influencer.”

But changing items can change how the measure works.

Document what you changed.

Check reliability again in your own sample.

Be cautious about claiming that the adapted version performs exactly like the original.

Established measure $\neq$ permission to stop thinking.

Mini literature-to-measure exercise

Your RQ: Is perceived sincerity of BrewBuddies' CEO apology related to trust in BrewBuddies?

1. Identify the two variables.
2. Which variable(s) would require a survey measure?
3. What search terms would you try in Google Scholar?
4. What would you look for in the Methods section?
5. What reliability information might you record?
6. What would make you question whether the measure fits your study?

Now that you learn about sampling and measurement (including reliability and validity), how do they work together?

- Strong measurement with a poor sample can limit who your findings represent.
- A strong sample with poor measurement can produce misleading data.
- Research quality depends on both WHO provides the data and HOW the variables are measured.
- These decisions should follow from the research question—not from convenience alone.

Good evidence = thoughtful sampling + thoughtful measurement.

Before collecting survey data, ask...

ABOUT THE SAMPLE

Who is my population? How will I recruit? Who might be missing? What can I reasonably generalize to?

ABOUT THE MEASURE

What exactly is my construct? How has it been measured before? Is there reliability/validity evidence? Does it fit my context?

Quiz

Answer these without looking back.

1. Population vs. sample: what is the difference?
3. Give one probability and one nonprobability sampling method.
4. What does reliability mean?
5. What does Cronbach's α tell us?
6. Why can a reliable measure still be invalid?